

STANLEY BARNES

Embedded Systems | FPGA & PCB Design | Edge Computing
Knoxville, TN • wstanleyb@gmail.com • wstanleyb.com

EDUCATION

B.S. Computer Engineering, University of Tennessee, Knoxville

Expected May 2027

GPA 3.92 / 4.00

Relevant Coursework: VLSI, System on Chip, Computer Architecture, Digital Systems Design, Embedded Systems, Microwave Circuits

EXPERIENCE

Undergraduate Research Assistant

May 2025 – Present

EPIC Lab — Min H. Kao Dept. of Electrical Engineering & Computer Science, University of Tennessee, Knoxville

- Designed, layed out, and validated custom PCBs for embedded sensing, power delivery, and edge computing;
- Developed firmware in C for ARM Cortex-M microcontrollers, integrated sensor peripherals over SPI, I²C, and UART with interrupt and DMA-driven data paths.
- Debugged hardware and firmware at the bench using oscilloscopes, logic analyzers, and spectrum analyzers to isolate signal-integrity, timing, and power faults.
- Collaborated across electrical, mechanical, and software boundaries under Dr. Sai Swaminathan; machined and 3D printed enclosures for field-deployed prototypes; research is currently under submission to ACM CHI 2027.

Teaching Assistant — COSC 559, Human-Computer Interaction

Aug 2026 – Present

University of Tennessee, Knoxville

- Lead weekly lab sections for 23 students, explaining technical material to mixed-experience audiences; graded programming assignments.

PROJECTS

Teachable Things — Handheld Edge AI Platform · STM32, C, LVGL, KiCad

Research Project

- Built a low-power handheld device that lets non-technical users build, train, and deploy custom AI models entirely on device.
- Firmware runs bare-metal on an STM32 within a **3 MB memory and 5 W power budget**, requiring careful management of DMA transfers, interrupt latency, and peripheral bandwidth.
- Implemented an LVGL touchscreen UI over i8080 display interface and designed the supporting multi-layer PCB in KiCad.

Smokey Buoy — Autonomous Water Quality Monitoring · KiCad, TFLite, LTspice, Fusion 360

Research Project

- Designed a low-cost, energy-efficient autonomous buoy that detects nitrates, phosphates, heavy metals, and industrial contaminants in the field; deployed with rural communities from Tennessee to Hawaii.
- Paired colorimetric assay strips with an onboard camera and edge AI inference to estimate pollutant concentration at **96% confidence** without a network connection — image capture, processing, and classification handled entirely on device.
- Engineered power delivery for continuous unattended solar operation; validated the analog front end in LTspice before fabrication.

Prosthetic 3D printed hand · Verilog

Junior Capstone

- Designed prosthetic hand capable of easily being printed and constructed for rapid adoption and prototyping
- Implemented control logic in Verilog based off user motion to activate individual joints

TECHNICAL SKILLS

Firmware / Embedded: C, C++, bare-metal development, STM32 (ARM Cortex-M), interrupts, DMA pipelines, SPI, I²C, UART, ADC/DAC, GPIO, timers

FPGA / Digital Design: Verilog, Xilinx FPGA toolchain, RTL design and simulation, clock-domain synchronization, finite state machines, timing analysis

Hardware Design: KiCad, Cadence Virtuoso, LTspice, Keysight ADS, Fusion 360, soldering/rework, 3D printing

Systems / Software: Linux, Windows, Mac, Python, Java, C/C++, Git, LVGL, TensorFlow Lite

Bench & Test: Oscilloscope, logic analyzer, waveform generator, spectrum analyzer

PUBLICATIONS

- Co-author, “Water is Life: Toward Open Infrastructures for Water Security Partnership,” *Conference on Human Factors in Computing Systems*, under review, August 2026
- Co-author, “Teachable Things: Interactive Machine Learning on Tiny Devices,” *Symposium on User Interface Software and Technology*, under review, June 2026